



Roll Number

NARULA INSTITUTE OF TECHNOLOGY
AN AUTONOMOUS INSTITUTE UNDER MAKAUT
2025
2024-25 EVEN SEMESTER
HU204 - ENVIRONMENTAL SCIENCE

TIME ALLOTTED: 3Hours**Maximum Marks : 70****Instructions to the candidate:***Figures to the right indicate full marks.**Draw neat sketches and diagram wherever is necessary.**Candidates are required to give their answers in their own words as far as practicable***Part A****Answer any ten from the following, choosing the correct alternative of each question: 10×1=10**

1. Ozone depletion can cause- (1) C02 BL2 PO1
(a) Skin cancer (b) Asthma
(c) Cholera (d) Fluorosis
2. Biogas contains- (1) C01 BL2 PO1
(a) CH₄ (b) CO₂
(c) N₂ and H₂S (d) All of these
3. Which of the following is not an example of man-made disaster- (1) C02 BL2 PO1
(a) Terrorism (b) Earthquake
(c) Toxic waste disposal (d) Oil spills
4. Logistic growth curve looks like as- (1) C02 BL2 PO1
(a) "S" shape (b) "Z" shape
(c) "J" shape (d) "L" shape
5. Blackfoot disease is related to- (1) C01 BL2 PO2
(a) Hg (b) As
(c) Pb (d) Cd
6. Temporary hardness of water is due to the presence of (1) C04 BL2 PO1
(a) Cl- (b) SO₄²⁻

- (c) PO4-3 (d) HCO3-
7. BOD test is done in dark to (1) C04 BL3 PO1
- (a) Prevent light to go inside the bottle (b) Prevent gas to defuse the bottle
- (c) Prevent light to go out (d) None of these
8. Synaecology can also be termed as (1) C01 BL1 PO1
- a) population ecology
- b) community ecology
- c) landscape ecology
- d) none of these
9. Hazardous waste should have the characteristics as (1) C04 BL3 PO1
- a) toxicity
- b) corrosivity
- c) reactivity
- d) all of these
10. Which the following contains maximum quantity of fresh water? (1) C01 BL1 PO1
- (a) Glaciers and Ice traps (b) River
- (c) Lakes (d) Soil Moisture
11. The example of secondary consumer (1) C01 BL1 PO1
- (a) Plant (b) Cow
- (c) Tiger (d) Frog
12. The primary pollutants responsible for photochemical smog formation are (1) C04 BL2 PO1
- (a) SO₂ and NO₂ (b) HC and NO₂
- (c) PAN and HC (d) Aldehyde and PAN

Part B**(Answer any three of the following) 3x5=15**

13. Answer the following:
- (a) What is equivalent noise level (Leq)? (1) C02 BL2 PO1

- (b) The measured noise in an area is 70 dB for 2 hrs, 90 dB for 45 minutes, 100 dB for 30 minutes, 110 dB for 10 minutes. Find out the noise threshold limit value and predict wheather the working environment is healthy or not? (4) C01 BL2 PO1
14. What is biogeochemical cycle? Discuss one of them briefly. (5) C02 BL1 PO1
15. a) How can solid wastes be disposed? (2) (5) C03 BL4 PO1
b) Explain briefly the method of composting. (3)
16. Classify the food chain with suitable examples. (5) C02 BL2 PO1
17. What is Maximum sustainable Yield? Prove that Maximum sustainable Yield is achieved when $N=K/2$, where the notations have their own meaning. (5) C01 BL3 PO1

Part C**(Answer any three of the following) 3x15=45**

18. Answer the following:
- (a) Briefly discuss the stages of disaster management cycle. (5) C01 BL3 PO2
- (b) Write down the objectives of Environmental Impact Assessment. (4) C02 BL3 PO2
- (c) How does sanitary landfilling differ from open dumping? (4) C02 BL4 PO2
- (d) Mention two air pollution controlling device. (2) C02 BL2 PO1
19. (a) What is acid rain? What are the adverse effects of acid rain on ecosystem and human civilization? (5) C04 BL2 PO1
- (b) What are the bio chemical effects of As^{3+} and Pb^{2+} ? (5) C03 BL2 PO1
- (c) Define BOD_5 . Explain the termsinvolved. What will the ratio of BOD_5 at 20 °C, to that of $BOD_{2.5}$ at 35 °C? (5) C04 BL3 PO1
20. (a) What is environmental management? What are the main components of environmental management? (5) C02 BL2 PO1
- (b) What is maximum sustainable yield (MSY)? Show that $(dn/dt)_{max} = R_0.K^2/4(K-N_0)$, the terms have their usual meaning (5) C04 BL3 PO1
- (c) What is biomagnification? What is the impact of biomagnifications on environment? (5) C02 BL2 PO1

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| 21. (a) | What is maximum mixing depth and ventilation coefficient?
How does atmospheric stability depend on them? | (5) | C04 | BL2 | PO1 |
| (b) | What is environmental management? What are the main
components of environmental management? | (5) | C02 | BL2 | PO1 |
| (c) | Briefly describe a grassland ecosystem with all of its
components. | (5) | C01 | BL2 | PO1 |
| 22. (a) | What are the different types of halocarbons? Give
examples. Give the formula for CFC-115 | (5) | C03 | BL3 | PO1 |
| (b) | Explain bio-magnifications. What is its significance in food
chain? | (5) | C01 | BL1 | PO1 |
| (c) | Give a brief account of earthquake management | (5) | C03 | BL2 | PO1 |